Visual Alignment Report

Methodology

The visual alignment pipeline (`scripts/18_visual_alignment.py`) evaluates structural preservation by directly comparing the layout of the reference PDF and the candidate PDF.

1. **Text Bounding Boxes (IoU)**: For text-heavy categories, PyMuPDF extracts bounding boxes for all text blocks. The coordinates are normalized by page size, matched primarily by reading order (with a Jaccard text-content fallback if block counts mismatch heavily), and scored via Intersection over Union (IoU).
2. **Structural Similarity (SSIM)**: For graphics-only categories (`tikz`, `pgfplots`, `posters`) without reliable text layers, pages are rendered to images, size-padded with white space to match dimensions, and evaluated via SSIM.
3. **Text Content Match (Jaccard)**: To ensure blocks are actually matching the right content, we compute a Jaccard similarity score for text tokens. We normalize Unicode (converting mathematical-italic letters to plain ASCII via NFKD) and strip punctuation (`re.findall(r'[a-z0-9]+')`) to accurately compare words. Page-number footer blocks (bottom 20% of the page, digits only) are filtered out on both sides before matching.

Categorical Thresholds

- **IoU (Text Position)**: `exact` (>= 0.70), `minor` (0.40 – 0.70), `major` (< 0.40)
- **SSIM (Graphics)**: `exact` (>= 0.95), `minor` (0.75 – 0.95), `major` (< 0.75)
- **Text Content Match (Jaccard)**: Tokens must have a similarity > 0.30 to be considered matching.

  IMPORTANT — Failed generations: If a model or engine fails to compile, fails to generate a PDF, or produces an empty/invalid file, it receives a strict `0.0` for both Position (IoU/SSIM) and Text Content Match. These zeroes **are included** in all categorical averages to accurately penalize unreliability.

  Provenance note: every score in this report is read from `ai_models/absolute_scores.json` as produced by the canonical pipeline run; the aggregate tables below are printed by `scripts/21_report_tables.py`, and every grid image embeds its Match/Pos labels from the same JSON at render time (`scripts/19_generate_report_assets.py`), so figures and tables cannot silently disagree.

Aggregated Averages (Alignment Only)

Fair-Subset Averages (AI Models vs Engines)

Averages calculated *only* over the subset of samples completed by the AI Models (8 segments).

Candidate	Text Content Match	Text Position (Mean IoU)
gemini	0.838	0.447
gpt	0.704	0.288
pandoc	0.642	0.230
tylax	0.776	0.281
typetex	0.713	0.243

NOTE — change from earlier drafts: previously published fair-subset numbers (e.g. Gemini 0.683/0.372) predate two matcher fixes: NFKD unicode normalization and page-number footer filtering. Both fixes raise most candidates' content-match rates (footers no longer count as unmatched blocks) and shift IoU values. GPT's two compile failures (`algorithms_easy`, `prose_hard`) are included as 0.0 in its averages.

WARNING — Block-Chunking Bias: The text alignment metric (Bounding Box IoU) structurally penalizes AI candidates for chunking layout content into fewer, larger blocks than the reference LaTeX/Pandoc engines (which emit heavily fragmented line-by-line blocks). A visually pristine

AI output can score mathematically lower than an equivalent engine output. Even so, Gemini now leads both content match and position on this subset.

Fair Subset Visual Comparisons (AI Models vs Engines)

Below are 3x2 grids comparing the reference layout to all 5 candidates for some of the 8 samples evaluated by the AI models. Panel labels show each candidate's Match / Pos scores from the scores table; failed candidates show a FAILED panel.

Sample: `tables_complex_hard`

Figure 1: Grid tables_complex_hard

Sample: `prose_easy`

Figure 2: Grid prose_easy

Sample: `eq_hard_hard`

Figure 3: Grid eq_hard_hard

Sample: `algorithms_easy`

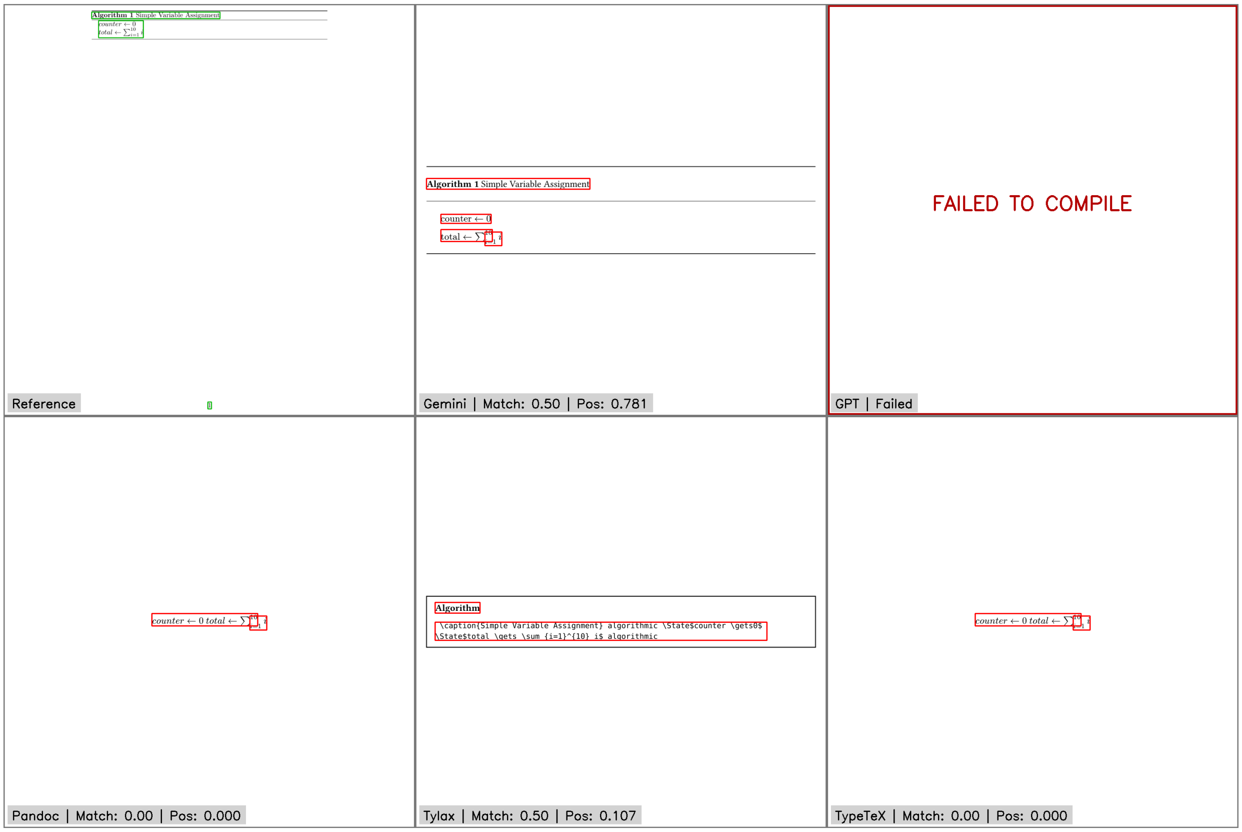

Figure 4: Grid algorithms_easy

Curated Examples: Content vs Position Trade-offs

These examples illustrate how the two metrics move independently.

Simple prose: both content and position can be near-perfect

Sample: `prose_easy` | **Candidate:** `gpt` | **Match Rate:** `1.00` | **IoU:** `0.704`

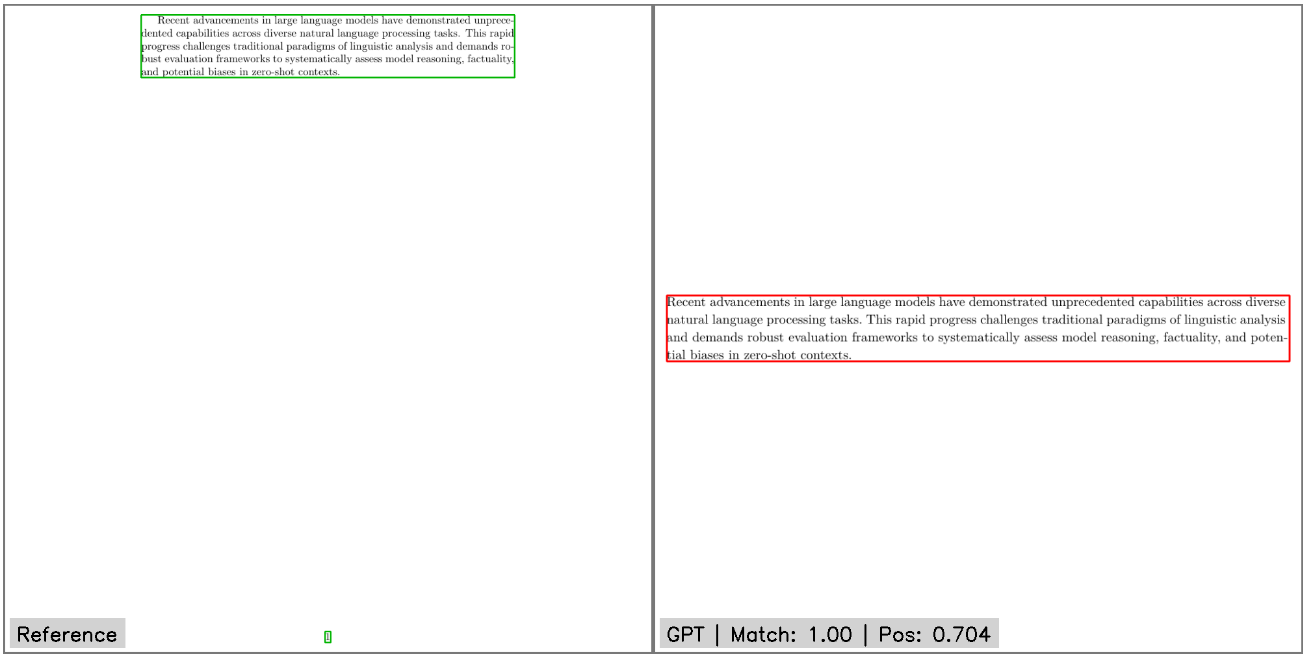

Figure 5: Curated Example 1

On single-page prose, GPT reproduces both the text and its layout closely enough to score `exact` (IoU 0.704). Earlier drafts used this sample to illustrate a low position score; after the NFKD and page-number-filter fixes, its position score is genuinely high.

High Content Match, Moderate Position Score (Table Formatting Shift)

Sample: `tables_complex_hard` | **Candidate:** `tylax` | **Match Rate:** `1.00` | **IoU:** `0.537`

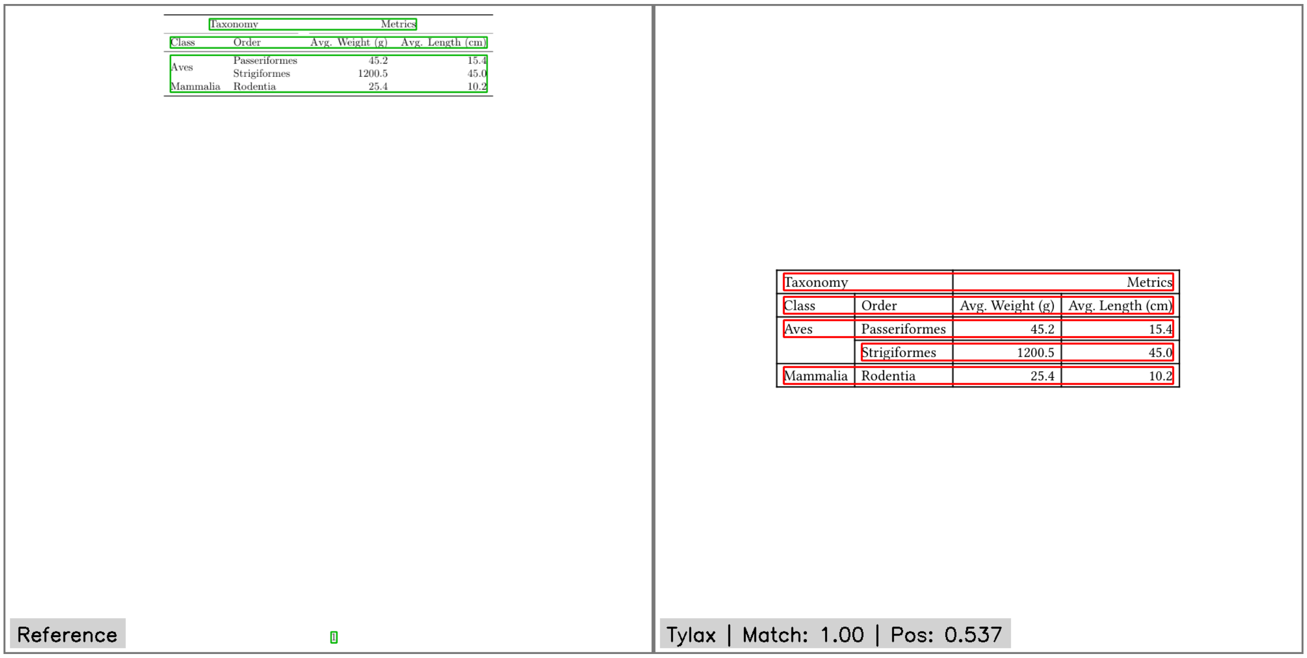

Figure 6: Curated Example 2

Tylax renders the table and its text is fully matched (1.00). However, the spatial dimensions and column widths of the generated table differ from the reference PDF, leading to a moderate position IoU (0.537) despite the content being intact.

The Impact of Patching CVs

Sample: `cv_complex_hard` | **Candidate:** `tylax_patched` | **Match Rate:** `1.00` | **IoU:** `0.000`

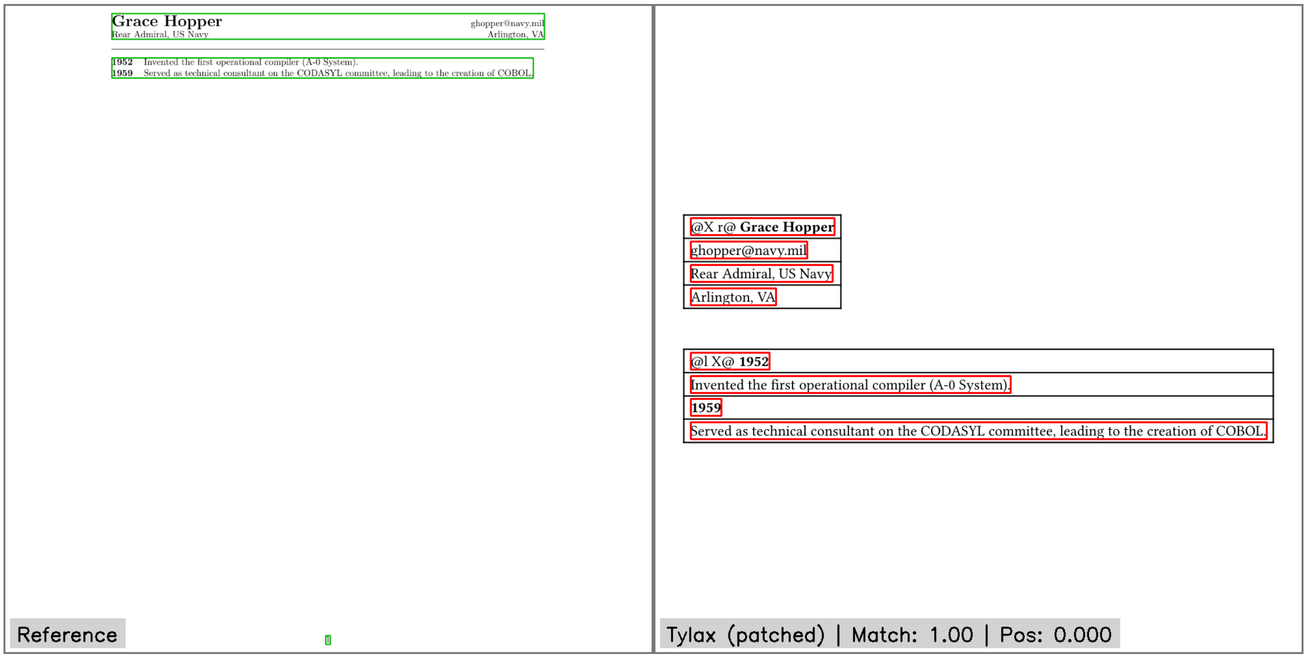

Figure 7: Curated Example 3

Tylax failed to compile this CV natively (unescaped `@`). After patching, it compiles and all text content is recovered (match 1.00), but the layout is so drastically rearranged that no matched block pair overlaps at all — position IoU 0.0. Patching restores content, not geometry.

Pandoc's Equation Handling

Sample: `eq_simple_hard` | **Candidate:** `pandoc` | **Match Rate:** `0.83` | **IoU:** `0.178`

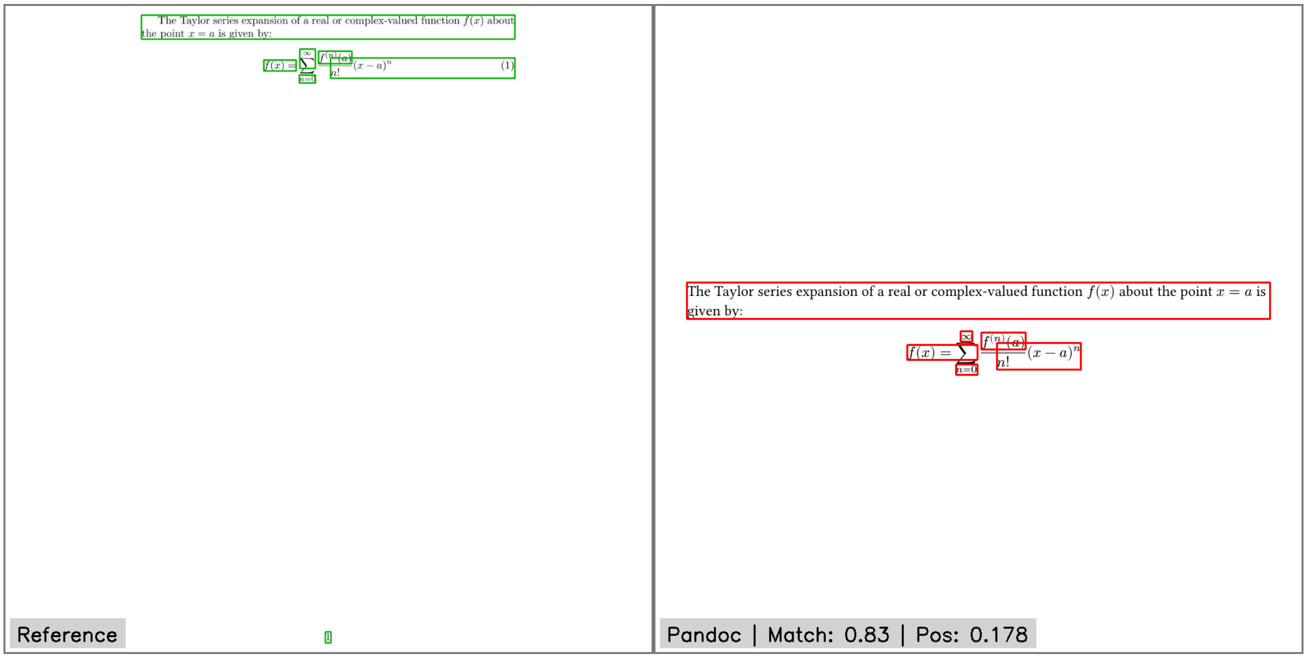

Figure 8: Curated Example 4

Pandoc outputs Unicode Math characters (mathematical italics) instead of styled ASCII; NFKD normalization lets these match the reference words fairly (match 0.83). The equations themselves land in visibly different vertical positions, so the position score stays low (0.178).

Full-Corpus Averages (Engines Only)

Averages calculated over the entire evaluation dataset of **48 samples**. Compile failures are included as 0.0 in all averages. Text Content Match averages over all 48 samples (a compiled graphics sample counts 1.0, since whole-image SSIM implies the full structure is present); Text Position averages over the 36 text samples; Graphics Alignment averages over the 12 graphics samples.

Candidate	Text Content Match	Text Position (Mean IoU)	Graphics Alignment (Mean SSIM)
pandoc	0.679	0.195	0.001
tylax	0.586	0.211	0.061
typetex	0.691	0.199	0.001

> **NOTE:** The near-zero SSIM values are expected: none of the three engines can translate TikZ/PGFPlots drawing commands, so the "graphics" pages compile as (near-)blank pages, and blank candidate pages are scored 0.0 by design. Tylax's nonzero SSIM comes from partial TikZ output, dominated by `tikz_simple_medium` (SSIM 0.697).

Note on "Patched" Candidates

Some candidate engines failed to compile CV formats natively and were manually patched (appearing as `pandoc_patched`, `tylax_patched`, and `typetex_patched`; TypeTeX's patched artifacts are stored as `typetex_approx_patched.*` on disk).

> **NOTE — Absence of Patched SSIM:** The patched variants are excluded from the visual alignment table above. The manual patches were applied only to CV samples (which contain no graphics and receive no SSIM scores), so a raw average for the patched variants would yield a mechanical `0.000` SSIM purely from an absence of graphical samples. The meaningful structural comparison for patched candidates is in `benchmark_findings.md`.

Visual Examples

Positive Examples (High Alignment Score)

Sample: `prose_easy` | **Candidate**: `pandoc` | **IoU Score**: `0.738` | **Match Rate**: `1.00`

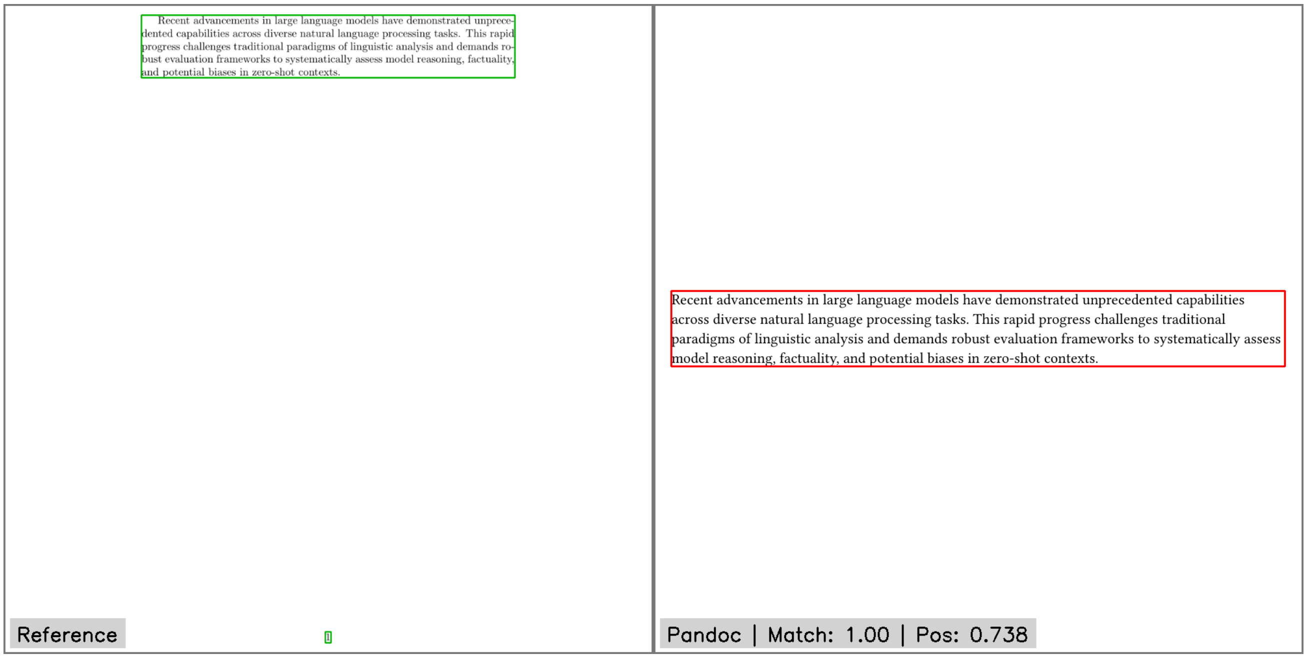

Figure 9: Positive Example 1

Sample: eq_simple_easy | **Candidate**: pandoc | **IoU Score**: 0.654 | **Match Rate**: 1.00

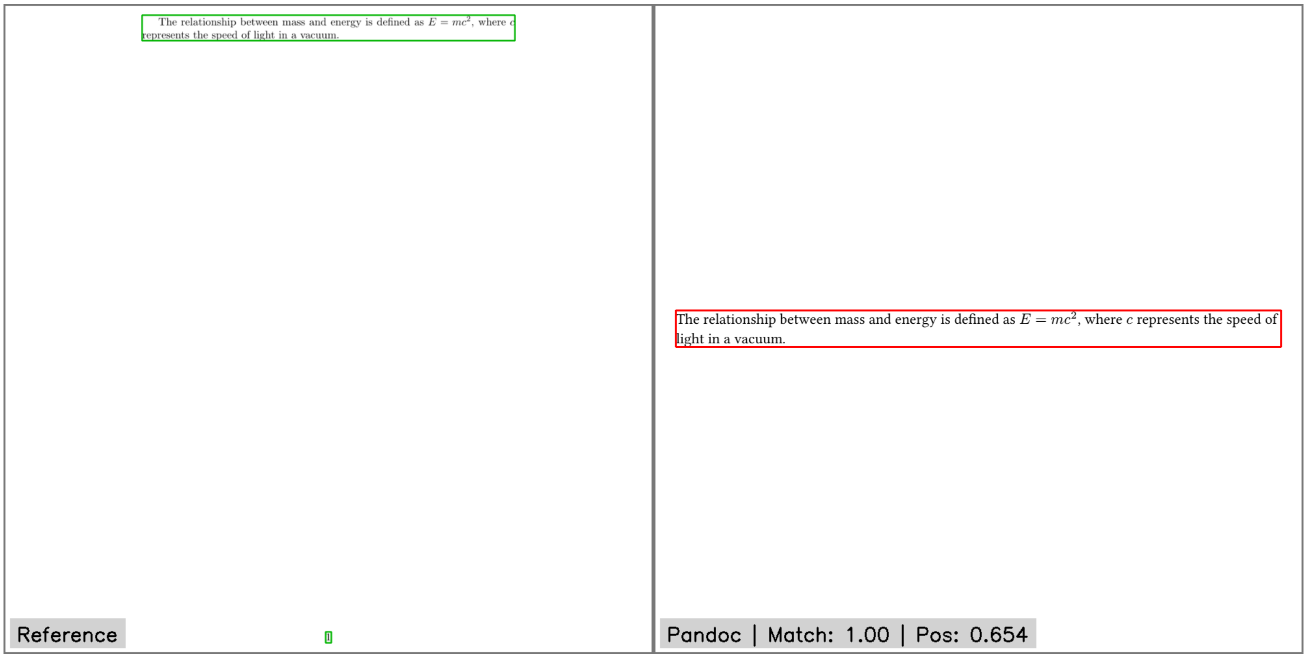

Figure 10: Positive Example 2

Negative Examples (Low Alignment Score)

Sample: `paper_full_easy` | **Candidate**: `typetex` | **IoU Score**: `0.000` | **Match Rate**: `0.22`

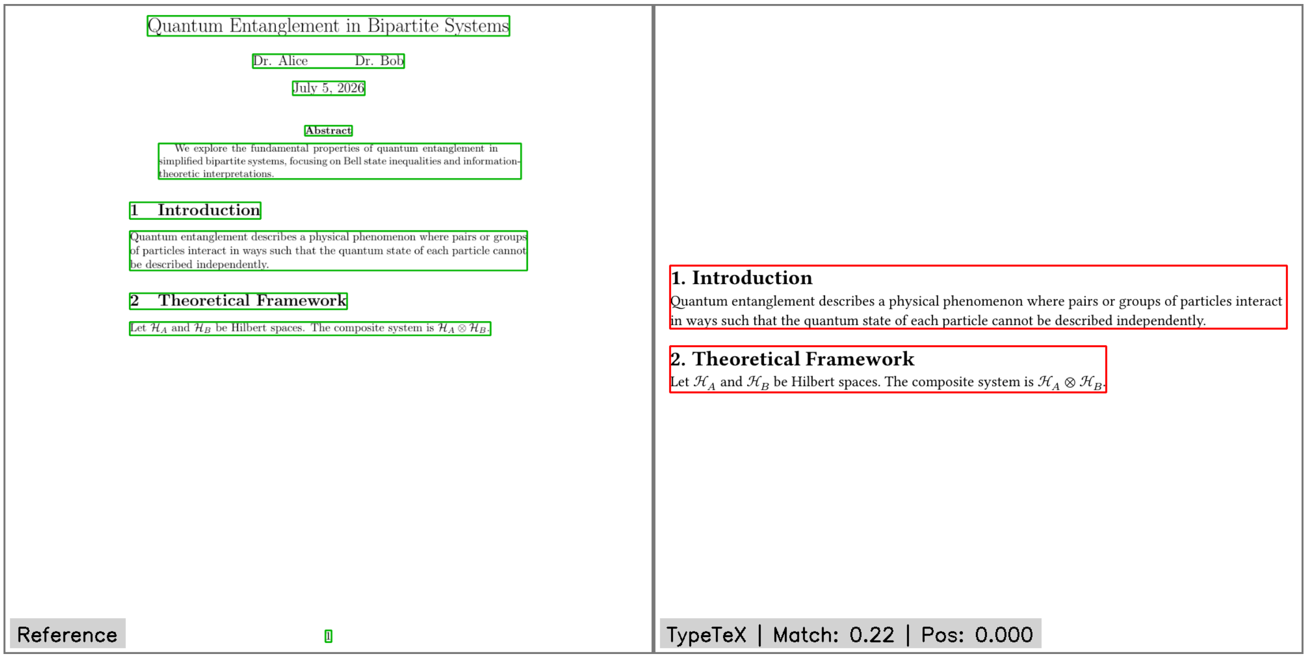

Figure 11: Negative Example 1

Sample: `tables_complex_attention_table2` | **Candidate**: `typetex` | **IoU Score**: `0.000` | **Match Rate**: `0.60`

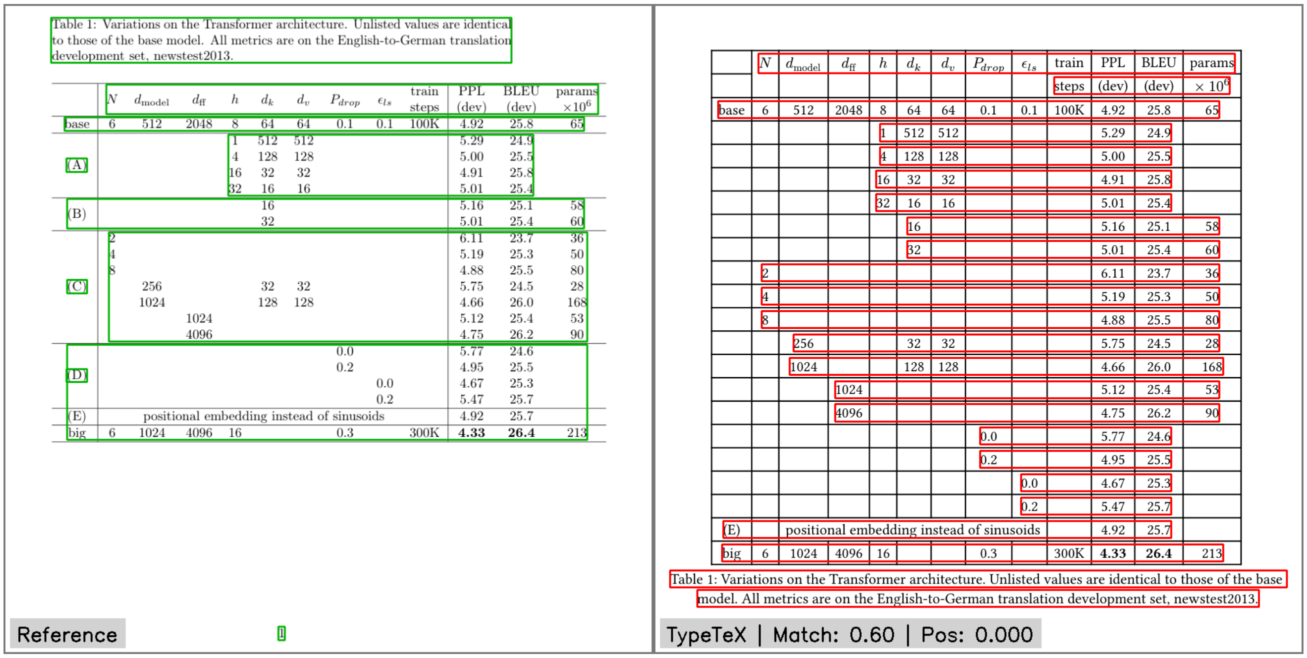

Figure 12: Negative Example 2

Limitations

While this visual comparison is much more robust than the previous heuristic, it does not capture: - Subtle font variations and kerning differences (unless they shift text blocks significantly). - Color mismatches (SSIM is computed in grayscale to focus on structure). - The actual reason for a mismatch (e.g. if a table overflows, the IoU score will just reflect the displacement). - Single-block references: when a reference PDF emits an entire table as one text block while the candidate emits per-row blocks (e.g. `tables_simple_easy`), no block pair can clear the Jaccard threshold and the sample scores a content mismatch even though the rendered text is visually identical. This is a known metric artifact and is left in the data rather than hand-corrected.

Appendix

The complete per-sample scores table and all 48 comparison grids are in `appendix.pdf`, generated by `scripts/20_generate_appendix.py`.

Full execution log of the scoring run used here: `logs/visual_alignment_20260709_220740.log`